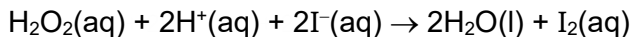


- 14 The reaction of hydrogen peroxide with iodide ions in an acidic solution can be monitored by an initial rates method.



The rate equation was found to be as follows:

$$\text{rate} = k [\text{H}_2\text{O}_2][\text{I}^-]$$

What of the following mechanism correctly describes this reaction?

- A** $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$ (slow)
 $\text{IO}^- + \text{H}^+ \rightarrow \text{HIO}$ (fast)
 $\text{HIO} + \text{H}^+ + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{I}_2$ (fast)
- B** $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$ (slow)
 $\text{H}_2\text{O}_2 + \text{IO}^- \rightarrow \text{H}_2\text{O} + \text{IO}_2^-$ (fast)
 $\text{IO}_2^- + \text{I}^- + 4\text{H}^+ \rightarrow 2\text{H}_2\text{O} + \text{I}_2$ (fast)
- C** $2\text{H}^+ + 2\text{I}^- \rightarrow 2\text{HI}$ (fast)
 $2\text{HI} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{I}_2$ (slow)
- D** $\text{H}_2\text{O}_2 + \text{I}^- + \text{H}^+ \rightarrow \text{H}_2\text{O} + \text{HIO}$ (fast)
 $\text{HIO} + \text{I}^- \rightarrow \text{OH}^- + \text{I}_2$ (slow)
 $\text{OH}^- + \text{H}^+ \rightarrow \text{H}_2\text{O}$ (fast)